Case-III :- H.W.

Lecture-7. Date 24.03.26.

§. Lagrange's Method for reduction of order of homogeneous linear ODEs.

$$y'' + p(x) y' + q(x) y = 0 \quad \text{--- 1)}.$$

If y_1 is a Solution of 1), then we can find another Solution y_2 of 1). Such that y_1, y_2 are linearly independent.

$:-$ $y_2 = u y_1,$

$$y_2' = u' y_1 + u y_1'.$$

$$y_2'' = u'' y_1 + 2u' y_1' + u y_1''$$

$$y_1 u'' + (2y_1' + p(x) y_1) u' + \underbrace{(y_1'' + p(x) y_1' + q(x) y_1)}_{= 0} u = 0$$

Let $U = u'$, $\quad y_1 \dfrac{dU}{dx} + (2y_1' + p(x) y_1) U = 0.$

$$\frac{du}{u} + \frac{2y_1' \, dx}{y_1} + p(x)\, dx = 0.$$

$$\log u + \log y_1^2 + \int p(x)\, dx = 0.$$

$$u = \frac{1}{y_1^2} e^{-\int p(x)\, dx}.$$

$$u' = \frac{1}{y_1^2} e^{-\int p(x) dx}.$$

$$u(x) = \int \frac{1}{y_1^2} e^{-\int p(x)\, dx}\, dx. \neq \text{constant.}$$

$y_2 = u(x) y_1$. ; So, y_1, y_2 are linearly independent.

§ Linear 2nd order non-homogeneous ODEs.

$$y'' + p(x) y' + q(x) y = r(x) \quad -1); \quad r(x) \neq 0.$$

$p(x)$, $q(x)$, and $r(x)$ are functions defined on $I \subseteq \mathbb{R}$.
interval.

$$y'' + p(x) y' + q(x) y = 0 \quad -2).$$

$$L := \_\_'' + p(x) \_\_' + q(x) \_\_$$

L is linear operator. i.e., $L(y_1) + L(y_2) = L(y_1 + y_2)$.
$L(cy_1) = c L(y_1)$.

Suppose y_p is a particular Solution of 1).
and y_1 is a Solution of 2).

Then $L(y_p+y_1) = L(y_p)+L(y_1) = r(x)+0 = r(x)$.

So, y_p+y_1 is a Solution of 1).

Let y be any solution of 1).

$$L(y-y_p) = L(y)-L(y_p) = r(x)-r(x) = 0.$$

$y-y_p$ is a Solution of 2).

$$y-y_p = c_1 y_1 + c_2 y_2 .$$

where y_1, y_2 are linearly independent.
c_1, c_2 are Constants.

$$y = c_1 y_1 + c_2 y_2 + y_p .$$

conversely, if $L(y_1) = 0 = L(y_2)$ and y_1, y_2 are linearly independent. and y_p is a particular Solution of 1), then

$$L(c_1 y_1 + c_2 y_2 + y_p) = r(x).$$

So, $y = c_1 y_1 + c_2 y_2 + y_p$ is a Solution of 1).

<u>Theorem</u>:- A general solution of (1) is a Solution of the form.

$$y = \underbrace{c_1 y_1 + c_2 y_2}+ y_p .$$
$$= y_h(x) + y_p(x).$$

where $y_h(x)$ is a general solution of (2).
and y_p is a particular solution of (1).

§ Lagrange's Method (Method of Variation of Parameters)

$$y_h = c_1 y_1 + c_2 y_2.$$

c_1, c_2 — constants.

y_1, y_2 are linearly independent.

Particular solution:

$$y_p = u_1 y_1 + u_2 y_2 , \quad u_1, u_2 \text{ are function of } x.$$

$$y_p' = u_1' y_1 + u_1 y_1' + u_2' y_2 + u_2 y_2'.$$

$$y_p'' = u_1'' y_1 + 2u_1' y_1' + y_1'' u_1 + u_2'' y_2 + 2u_2' y_2' + y_2'' u_2 .$$

$$\underbrace{L(y_1)}_{\substack{\parallel \\ 0}} u_1 + (2y_1' + p(x)y_1)u_1' + u_1'' y_1 + \underbrace{L(y_2)}_{\substack{\parallel \\ 0}} u_2 + (2y_2' + p(x)y_2)u_2' + u_2'' y_2 = r(x).$$

$$u_1'' y_1 + u_2'' y_2 + 2(y_1' u_1' + y_2' u_2') + p(x)\underbrace{(y_1 u_1' + y_2 u_2')} = r(x).$$

$$y_1 u_1' + y_2 u_2' = 0 \quad \text{—— eq 1)}.$$

$$y_1 u_1'' + y_2 u_2'' + y_1' u_1' + y_2' u_2' = 0.$$

$$y_1' u_1' + y_2' u_2' = r(x) \quad \text{— eq 2)}.$$

Since y_1, y_2 are linearly independent,

$W(y_1, y_2)(x) \neq 0,$

$$u_1' = \frac{\begin{vmatrix} 0 & y_2 \\ r(x) & y_2' \end{vmatrix}}{W(y_1,y_2)(x).} = \frac{-y_2 r(x).}{W(y_1,y_2)(x)}.$$

$$u_2' = \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & r(x) \end{vmatrix}}{W(y_1,y_2)(x)} = \frac{y_1 r(x).}{W(y_1,y_2)(x)}$$

$$u_1 = -\int \frac{y_2 r(x)\, dx.}{W(y_1,y_2)(x).}, \quad u_2 = \int \frac{y_1 r(x)}{W(y_1,y_2)(x).} dx.$$

$$y_p = u_1 y_1 + u_2 y_2 = y_2 \int \frac{y_1 r(x)\, dx.}{W(y_1,y_2)(x)} - y_1 \int \frac{y_2 r(x)\, dx.}{W(y_1,y_2)(x)}$$

General solution of (1) is.

$$y(x) = c_1 y_1 + c_2 y_2 + y_p.$$

<u>Example.</u> $4y'' - y = e^x.$

$$y'' - \frac{1}{4} y = \underbrace{\frac{1}{4} e^x}_{r(x).} \quad \text{— 1)}.$$

$$y'' - \frac{1}{4}y = 0 \qquad \text{—— } 2).$$

$y = e^{\lambda x}$.

$y'' = \lambda^2 e^{\lambda x}$.

$$(\lambda^2 - \frac{1}{4}) e^{\lambda x} = 0.$$

$$\Rightarrow \quad \lambda^2 - \frac{1}{4} = 0.$$

$$\Rightarrow \quad \lambda = \pm \frac{1}{2}.$$

$$y_1 = e^{x/2}, \quad y_2 = e^{-x/2}.$$

$$y_h = c_1 e^{x/2} + c_2 e^{-x/2}.$$

$$W(y_1, y_2) = \begin{vmatrix} e^{x/2} & e^{-x/2} \\ \frac{1}{2}e^{x/2} & -\frac{1}{2}e^{-x/2} \end{vmatrix} = -\frac{1}{2} - \frac{1}{2} = -1.$$

$$y_p = u_1 y_1 + u_2 y_2 \qquad \text{Eq } 1).$$

$$u_1' y_1 + u_2' y_2 = 0$$

$$u_1' y_1' + u_2' y_2' = \frac{1}{4} e^{x}.$$

$$u_1' = \frac{\begin{vmatrix} 0 & y_2 \\ \frac{1}{4}e^{x} & y_2' \end{vmatrix}}{-1} = \frac{\begin{vmatrix} 0 & e^{-x/2} \\ \frac{1}{4}e^{x} & -\frac{1}{2}e^{-x/2} \end{vmatrix}}{-1} = +\frac{1}{4} e^{x/2}.$$

$$u_1 = \frac{1}{2} e^{x/2}.$$

$$u_2' = \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & r(x) \end{vmatrix}}{-1} = \frac{\begin{vmatrix} e^{x/2} & 0 \\ \frac{1}{2}e^{x/2} & \frac{1}{4}e^{x} \end{vmatrix}}{-1} = -\frac{1}{4}e^{\frac{3x}{2}}.$$

$$u_2 = -\frac{1}{6}e^{\frac{3x}{2}}.$$

$$y_p = u_1y_1 + u_2y_2 = \frac{1}{2}e^{x/2}.e^{x/2} - \frac{1}{6}e^{3x/2}.e^{-x/2}$$
$$= \left(\frac{1}{2}-\frac{1}{6}\right)e^{x} = \frac{1}{3}e^{x}.$$

General solution of 1) is.

$$y = y_h + y_p.$$
$$= \underbrace{c_1e^{x/2} + c_2e^{-x/2}}_{y_h.} + \underbrace{\frac{1}{3}e^{x}}_{y_p}.$$

Lecture-8 Date 25.03.26.

Example. 1. $y'' + 6y' + 9y = 0$. —1)

Let $y = e^{\lambda x}$ be a trial solution.

$$(\lambda^2 + 6\lambda + 9)e^{\lambda x} = 0.$$
$$\Rightarrow \lambda^2 + 6\lambda + 9 = 0.$$
$$\Rightarrow (\lambda+3)^2 = 0. \qquad \lambda = -3,$$

$y_1 = e^{-3x}$. $\quad y_2 = x\,e^{-3x}$.

general solution.

$$y = c_1 e^{-3x} + c_2 x e^{-3x} = (c_1 + x c_2)\, e^{-3x}.$$

where c_1, c_2 are constants.

Example. $y'' - 2y' + y = 0$.

general Solution $y = (c_1 + c_2 x)e^{x}$.

Example. $ty'' - (4t+4)y' + (4t+8)y = 0 \quad \text{—1)} \; ; \quad t > 0$.

Given that $y_1(t) = e^{2t}$ is a Solution of 1).

find a Solution y_2, s.t y_1, y_2 are linearly indepent.

$= \frac{t^5}{5} e^{2t}$.

Example.: $(x^2 - x)y'' - xy' + y = 0$. —1).

Given that $y_1 = x$ is a Solution of 1)

Find a Solution y_2, Such that y_1, y_2 are linearly independent.

Example.: Solve the following IVP.

$$2y'' + y' - y = 0; \quad y(0) = 1, \; y'(0) = 2.$$

Ans: $\quad y(t) = 2e^{t/2} - e^{-t}$.

Example: $\quad y'' - y' - 2y = e^{-x}$.

$$y_h = c_1 e^{-x} + c_2 e^{2x}, \quad y_p \overset{?}{=} -\frac{(1+3x)}{9} e^{-x}.$$

$$u_1' y_1 + u_2' y_2 = 0 \quad — 1)$$

$$u_1' y_1' + u_2' y_2' = e^{-x} \quad — 2)$$

$$W(y_1, y_2) = \begin{vmatrix} e^{-x} & e^{2x} \\ -e^{-x} & 2e^{2x} \end{vmatrix} = +3e^{x}.$$

$$u_1' = \frac{\begin{vmatrix} 0 & y_2 \\ e^{-x} & y_2' \end{vmatrix}}{3e^{x}} = \frac{-y_2 e^{-x}}{3e^{x}} = -\frac{1}{3}$$

$$u_1 = -\frac{x}{3}.$$

$$u_2' = \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & e^{-x} \end{vmatrix}}{3e^{x}} = \frac{y_1 e^{-x}}{3e^{x}} = \frac{e^{-3x}}{3}.$$

$$u_2 = -\frac{1}{9} e^{-3x}.$$

$$y_p = u_1 y_1 + u_2 y_2 = -\frac{x}{3} e^{-x} - \frac{1}{9} e^{-3x} \cdot e^{2x} = \underline{-\left(\frac{x}{3} + \frac{1}{9}\right) e^{-x}}.$$

Example $\quad y'' + y = \sec x.$

$$y_p = x\sin x - \cos x \ln|\sec x|$$

$$y_h = c_1 \cos x + c_2 \sin x.$$

$$y = c_1 \cos x + c_2 \sin x + x\sin x - \cos x \ln|\sec x|.$$

* ——————————————————— *

Example: $y^{(4)} + y = 0$; $\lambda^4 + 1 = 0$.

$y_1 = e$ $\lambda = e^{\pi/4 i}, e^{\frac{3\pi}{4} i}, e^{-\frac{3\pi}{4} i}, e^{-\pi/4 i}$

$e^{\pi/4 i} = \cos\frac{\pi}{4} + i\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}} \pm i\frac{1}{\sqrt{2}}$.

$e^{\frac{3\pi}{4} i} = \cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4} = -\frac{1}{\sqrt{2}} \pm \frac{i}{\sqrt{2}}$.

$e^{\frac{1}{\sqrt{2}}x}\cos x\frac{1}{\sqrt{2}}$; $e^{1/\sqrt{2}\,x}\sin\frac{x}{\sqrt{2}}$. $(\lambda-1)^4 = 0$.

$e^{-\frac{1}{\sqrt{2}}x}\cos\left(\frac{1}{\sqrt{2}}x\right)$; $e^{-\frac{1}{\sqrt{2}}x}\sin\left(\frac{x}{\sqrt{2}}\right)$. $\underline{\underline{1, 1, 1, 1.}}$

$e^x, xe^x, x^2e^x, x^3e^3$.

$(\lambda^2+1)^2 = 0$.

$\lambda = \pm i$, $\lambda = \pm i$, $i \to 2$.
$-i \to 2$.

$i \to \cos x, \sin x$. $c_1\cos x + c_2\sin x$.
$x\cos x, x\sin x$. $+ c_3 x\cos x + c_4 x\sin x$.

Lecture-09 Date 27.03.26.

An expression of the form.

$$F(x, y, y^{(1)}, \ldots, y^{(n)}) = 0$$ is Called.

n-th order ODE.

An expression of the form.

$$y^{(n)} + p_{n-1}(x)\, y^{(n-1)} + p_{n-2}(x)\, y^{(n-2)} + \cdots + p_1(x)\, y^{(1)} + p_0(x)\, y^{(0)} = r(x). \quad (*)$$

is called n-th order linear ODE.

$$L(\_) = \_^{(n)} + p_{n-1}(x)\, \_^{(n-1)} + \cdots + p_1(x)\, \_^{(1)} + p_0(x)\, \_^{(0)}$$

linear operator:
- $L(y_1 + y_2) = L(y_1) + L(y_2)$.
- $L(c y_1) = c L(y_1)$.

$(*)$ is called <u>homogeneous</u> if $r(x) = 0$.
If $r(x) \neq 0$, $(*)$ is called <u>non-homogeneous</u>.

<u>Definition</u>: Let $\phi_1, \ldots, \phi_n$ are functions defined on an interval $I\ (\subseteq \mathbb{R})$
Then $\phi_1, \ldots, \phi_n$ are called <u>linearly independent</u>
if
$$c_1 \phi_1 + \cdots + c_n \phi_n = 0, \text{ then } c_1 = c_2 = \cdots = c_n = 0.$$
$$\left(\text{i.e., } c_1 \phi_1(x) + \cdots + c_n \phi_n(x) = 0 \quad \forall x \in I \right).$$

<u>IVP</u> (n-th order linear homogeneous ODEs).

$$L(y) = y^{(n)} + p_{n-1}(x)\, y^{(n-1)} + \cdots + p_1(x)\, y^{(1)} + p_0\, y^{(0)} = 0. \quad — 1)$$

where $p_{n-1}(x), \ldots, p_1(x), p_0(x)$ are (continuous) functions defined on some interval I containing x_0. $(*)$

$$y(x_0) = k_0,\ y^{(1)}(x_0) = k_1, \ldots, y^{(n-1)}(x_0) = k_{n-1}, \quad — 2)$$

where $k_0, \ldots, k_{n-1}$ are constants.

Theorem (Existence and uniqueness).

If $p_{n-1}(x), \ldots, p_1(x), p_0(x)$ are continuous functions defined on I containing x_0. Then. the above. IVP $(*)$ has a unique solution $y = y(x)$. defined on. I.

Remark: Any IVP has. a Solution.

Theorem.1 (Existence of n-linearly independent Solutions.).

There exist n- linearly independent. Solutions of $L(y)=0$. on I.

Proof: Sketch: $\quad y_i^{(i-1)}(x_0) = 1, \quad y_i^{(j-1)}(x_0) = 0; \quad i \neq j$
$1 \leq i, j \leq n$.

$$y_1(x_0) = 1,\ y_1^{(1)}(x_0) = 0,\ \ldots,\ y_1^{(n-1)}(x_0) = 0. \longrightarrow y_1 = y_1(x)$$
$$y_2(x_0) = 0,\ y_2^{(1)}(x_0) = 1,\ \ldots,\ y_2^{(n-1)}(x_0) = 0 \longrightarrow y_2 = y_2(x)$$

↳ **H.W.**

Theorem: Let $y_1, \ldots, y_n$ be the n-Solutions. of $L(y)=0$ on I Satisfying.

$$y_i^{(i-1)}(x_0) = 1 \quad \& \quad y_i^{(j-1)}(x_0) = 0 \text{ for all } i \neq j$$

If Φ is any Solution of $L(y)=0$, then.

$$\Phi = \sum c_i y_i, \quad c_1, \ldots, c_n \text{ are constants.}$$

Proof:- $\Phi(x_0)=\alpha_0,\ \Phi^{(1)}(x_0)=\alpha_1,\ldots,\ \Phi^{(n-1)}(x_0)=\alpha_{n-1}$

Take, $\Psi=\sum_{i=1}^{n}\alpha_{i-1}y_i$. Then.

$L(\Psi)=0,$ & $\Psi(x_0)=\alpha_0,\ \Psi'(x_0)=\alpha_1,\ldots,\Psi^{(n-1)}(x_0)=\alpha_{n-1}$.

Since Φ and Ψ are both solutions of $L(y)=0$, by the uniqueness theorem we have.

$$\Phi=\Psi=\sum_{i=1}^{n}\alpha_{i-1}y_i=\sum_{i=1}^{n}c_i y_i$$

constants.

Wronskian and linear independence:

$$L(y)=0 \quad\text{———}(1)$$

Let $y_1,\ldots,y_n$ be n- solutions of (1).
The Wronskian of $y_1,\ldots,y_n$ is denoted by by $W(y_1,\ldots,y_n)(x)$ and it is defined as

$$W(y_1,\ldots,y_n)(x)=\begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & & & \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}$$

Theorem:- Let $y_1,\ldots,y_n$ are solutions of (1) defined on some open interval I. Then $y_1,\ldots,y_n$ are linearly independant iff

$W(y_1, \ldots, y_n)(x) \neq 0 \;\; \forall\, x \in I.$

Remark:- $y_1, \ldots, y_n$ are linearly independent solutions of (1) iff $W(y_1, \ldots, y_n)(x_0) \neq 0$ for some $x_0 \in I$

Theorem*: Let $y_1, \ldots, y_n$ be n- linearly independent solutions of $L(y)=0$ defined on I.
If Φ is a solution of $L(y)=0$, then.

$$\Phi = c_1 y_1 + \cdots + c_n y_n, \quad c_1, \ldots, c_n \text{ are constants.}$$

§ n-th order linear homogeneous ODEs with constants coefficients.

$$y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y^{(1)} + a_0 y = 0. \quad \text{—(1)}$$

where. $y^{(n)} = \frac{d^n y}{dx^n}$.

Let $y = e^{\lambda x}$ be a trial solution. of (1).

$$y^{(i)} = \lambda^i e^{\lambda x}.$$

$$(\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1 \lambda + a_0)\, e^{\lambda x} = 0.$$

$$\Rightarrow \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1 \lambda + a_0 = 0. \quad \text{—(2)}$$

(2) is called the characteristic equation of (1)

Case-I. All the roots are real and distinct.

$\lambda = \lambda_1, \dots, \lambda_n.$

$y_i = e^{\lambda_i x}. \quad i = 1, \dots, n.$

$$W(y_1, \dots, y_n) = \begin{vmatrix} e^{\lambda_1 x} & e^{\lambda_2 x} & \cdots & e^{\lambda_n x} \\ \lambda_1 e^{\lambda_1 x} & \lambda_2 e^{\lambda_2 x} & \cdots & \lambda_n e^{\lambda_n x} \\ \vdots & & & \\ \lambda_1^{n-1} e^{\lambda_1 x} & \lambda_2^{n-1} e^{\lambda_2 x} & \cdots & \lambda_n^{n-1} e^{\lambda_n x} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & \cdots & 1 \\ \lambda_1 & \lambda_2 & & \lambda_n \\ \vdots & & & \\ \lambda_1^{n-1} & \lambda_2^{n-1} & \cdots & \lambda_n^{n-1} \end{vmatrix} e^{(\lambda_1 + \cdots + \lambda_n)x}.$$

Vandermonde determinant.

$$= \prod_{1 \le i < j \le n} (\lambda_j - \lambda_i) \cdot e^{(\lambda_1 + \cdots + \lambda_n)x}.$$

$$\neq 0. \quad [\lambda_i \neq \lambda_j \ \forall \ i \neq j]$$

The $y_1, \dots, y_n$ are linearly independent.

So the general solution is.

$$y = c_1 e^{\lambda_1 x} + \cdots + c_n e^{\lambda_n x}.$$